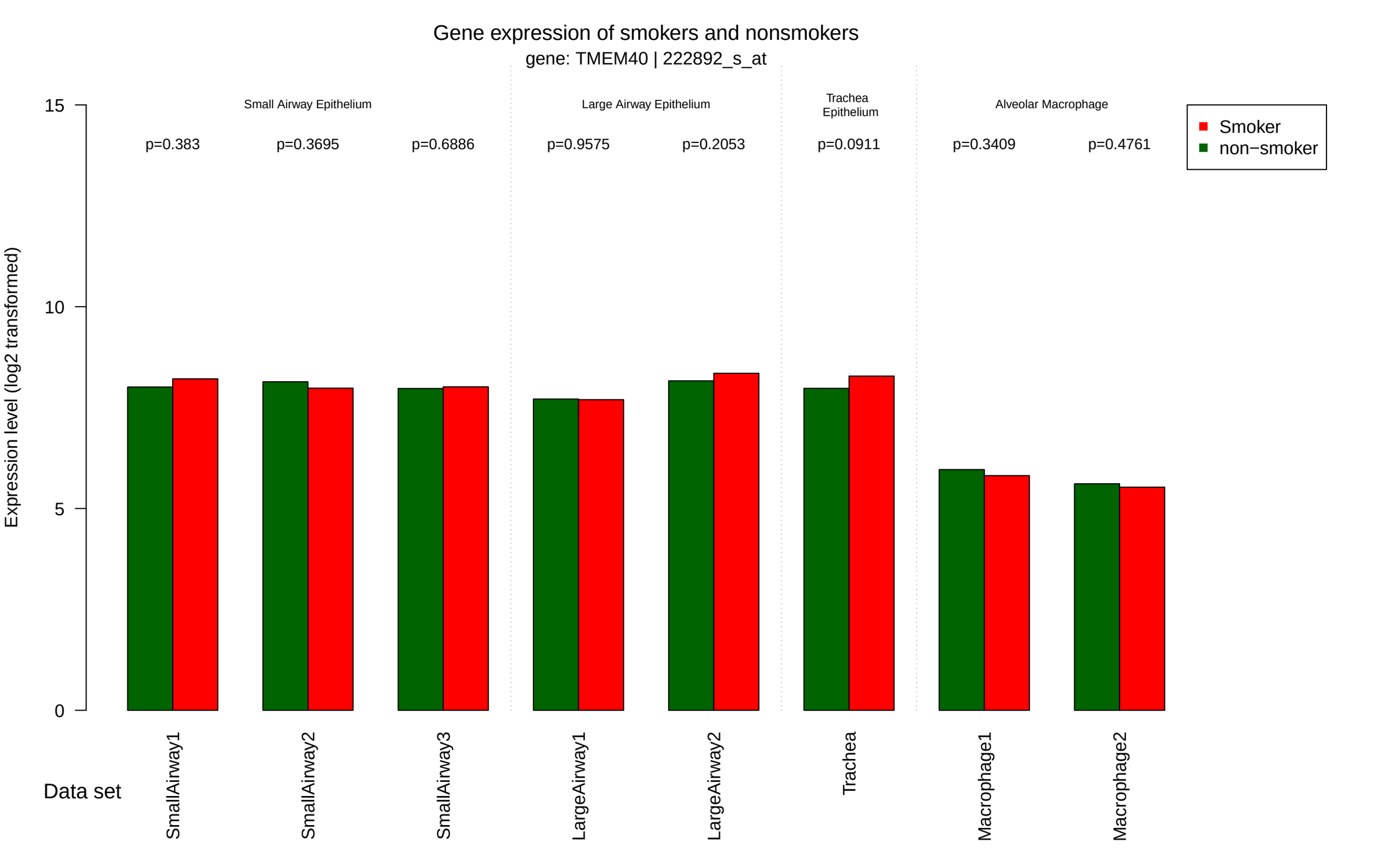
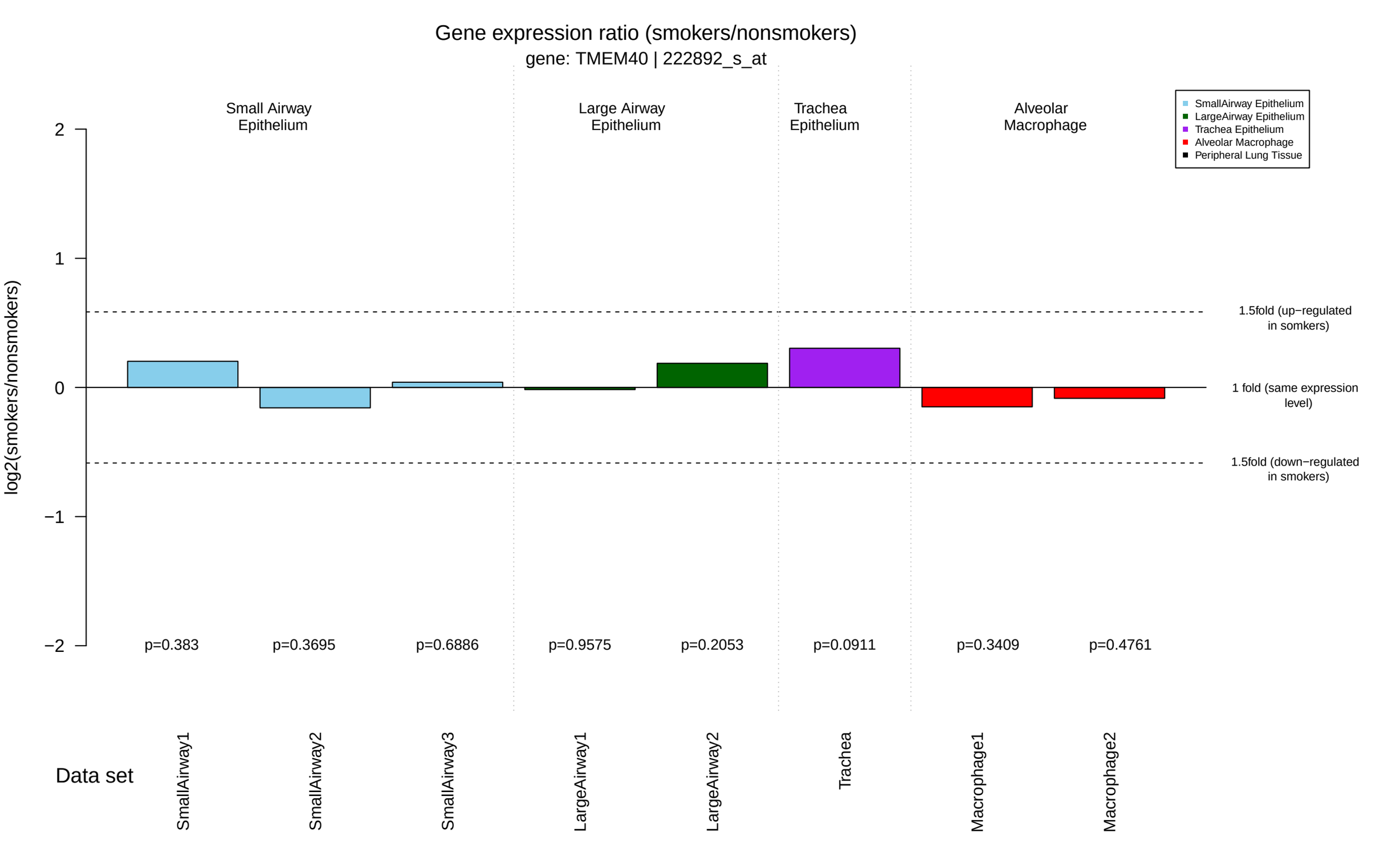
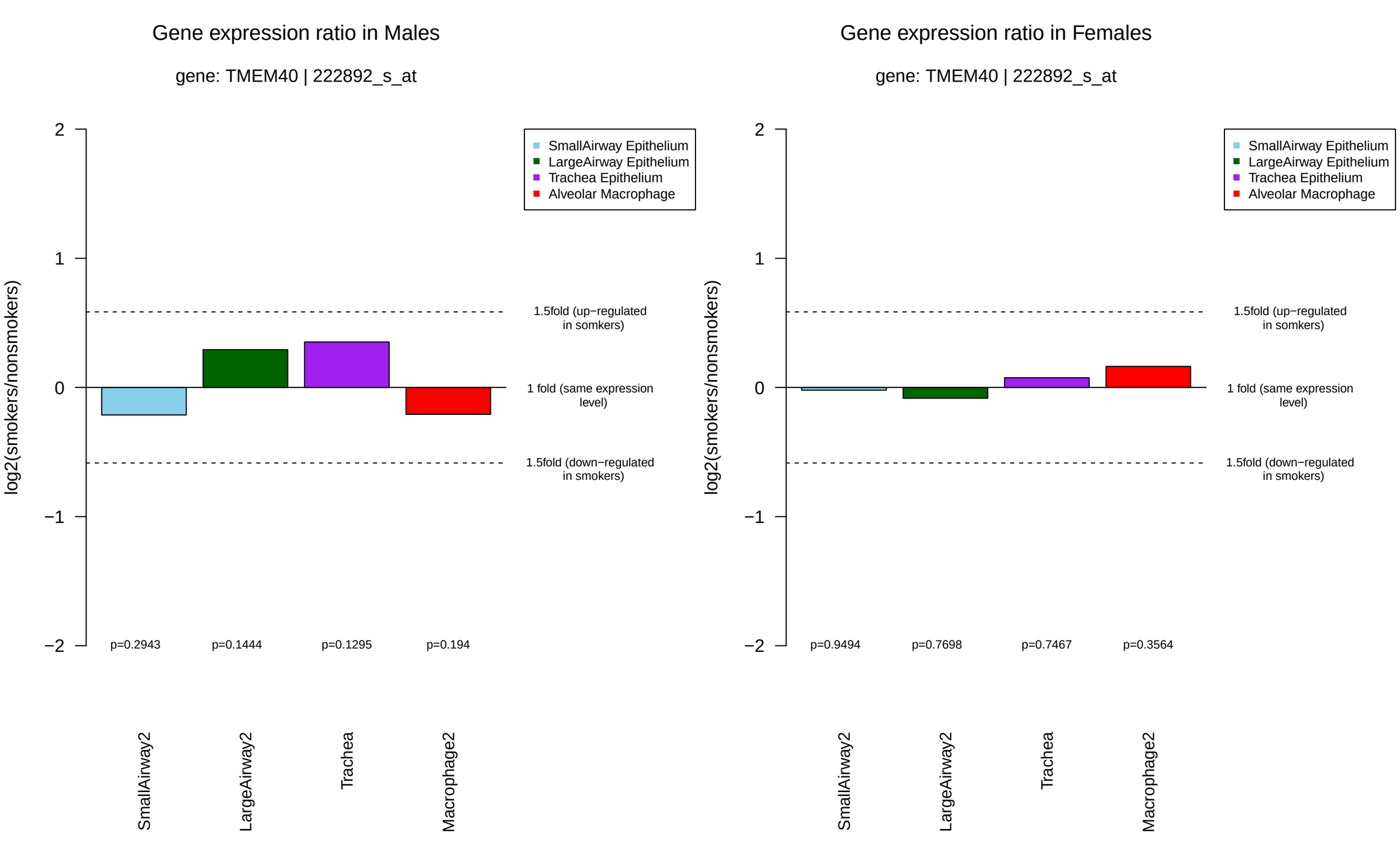
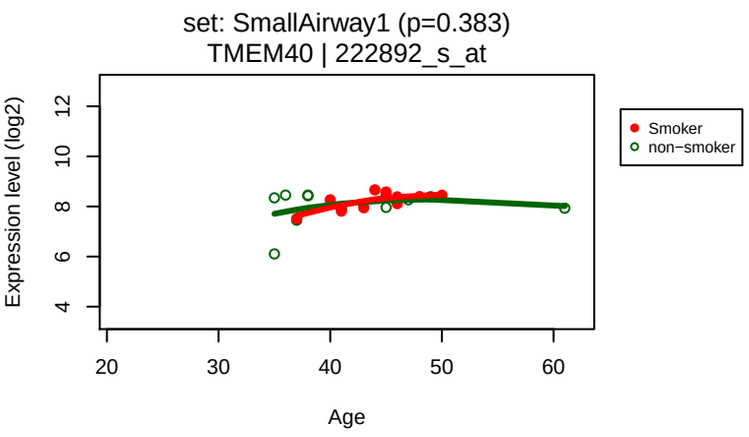
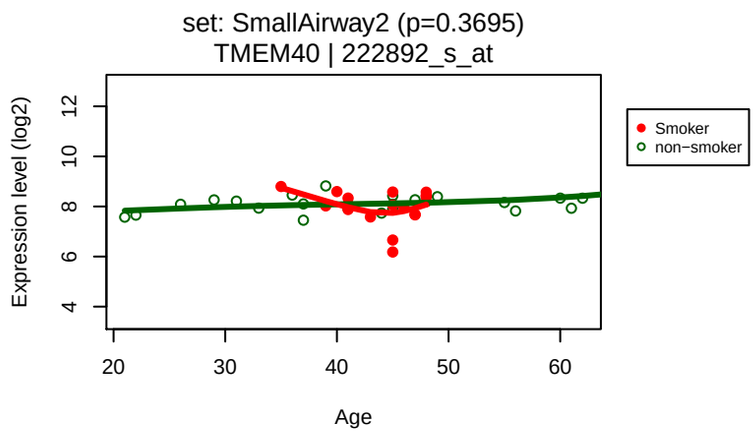
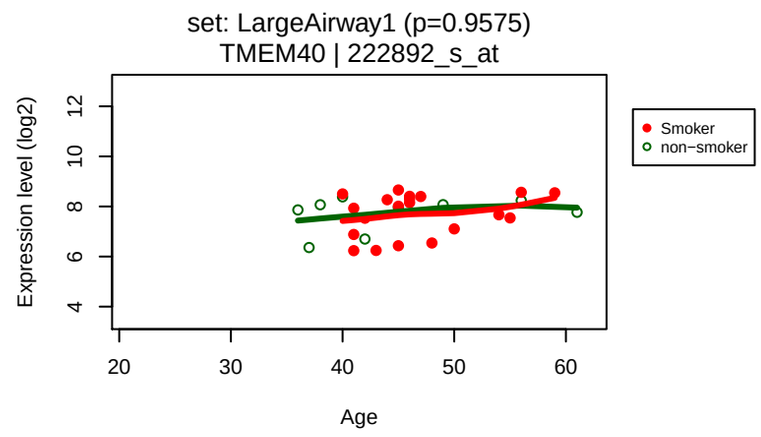

Gene expression of smokers and nonsmokers
gene: TMEM40 | 222892_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.383
p=0.3695
p=0.6886
p=0.9575
p=0.2053
p=0.0911
p=0.3409
p=0.4761
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

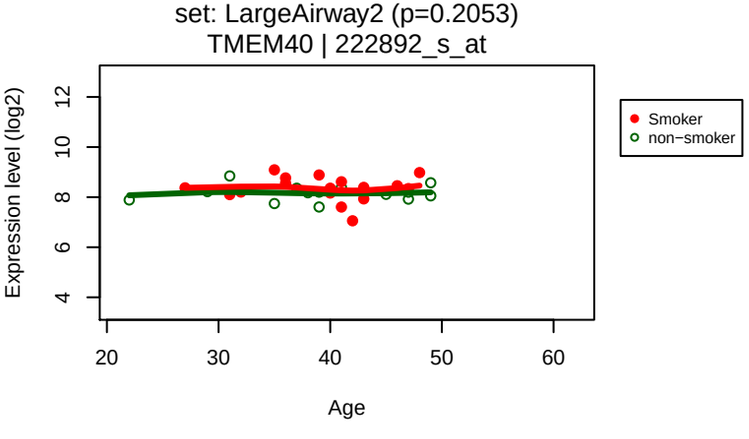

Gene expression ratio (smokers/nonsmokers)
gene: TMEM40 | 222892_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.383
p=0.3695
p=0.6886
p=0.9575
p=0.2053
p=0.0911
p=0.3409
p=0.4761
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

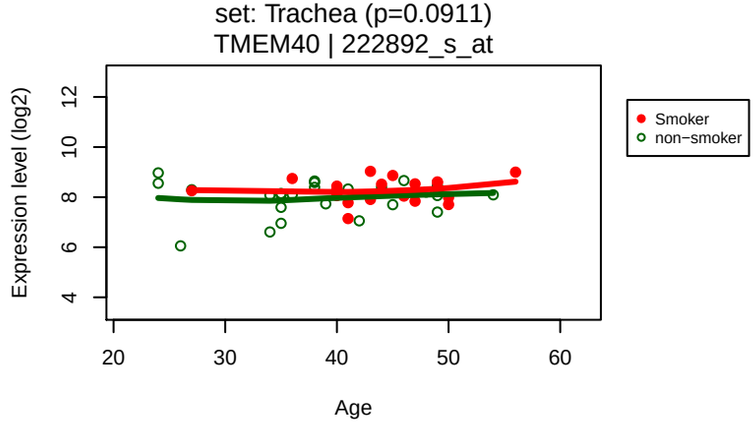

Gene expression ratio in Males
gene: TMEM40 | 222892_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2943
p=0.1444
p=0.1295
p=0.194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM40 | 222892_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9494
p=0.7698
p=0.7467
p=0.3564
SmallAirway2
LargeAirway2
Trachea
Macrophage2

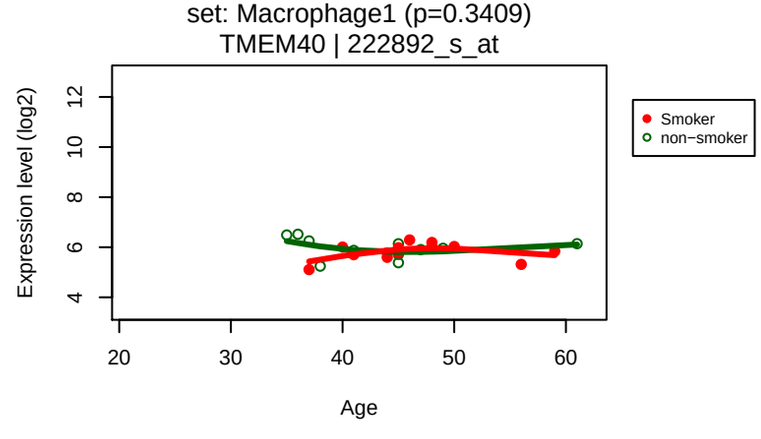

set: SmallAirway1 (p=0.383)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker

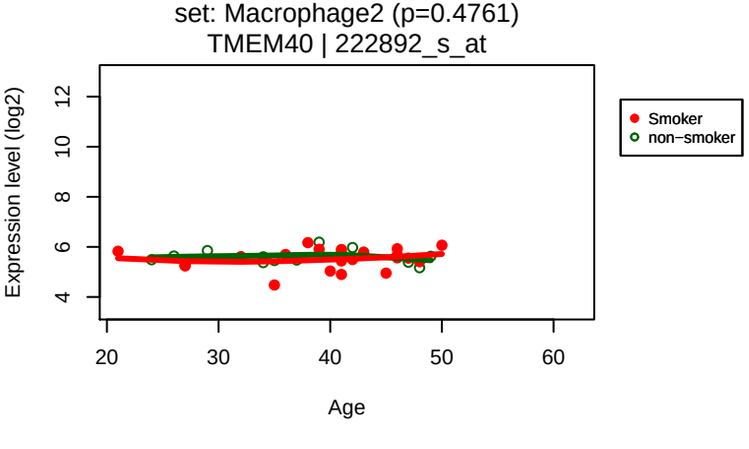

set: SmallAirway2 (p=0.3695)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker

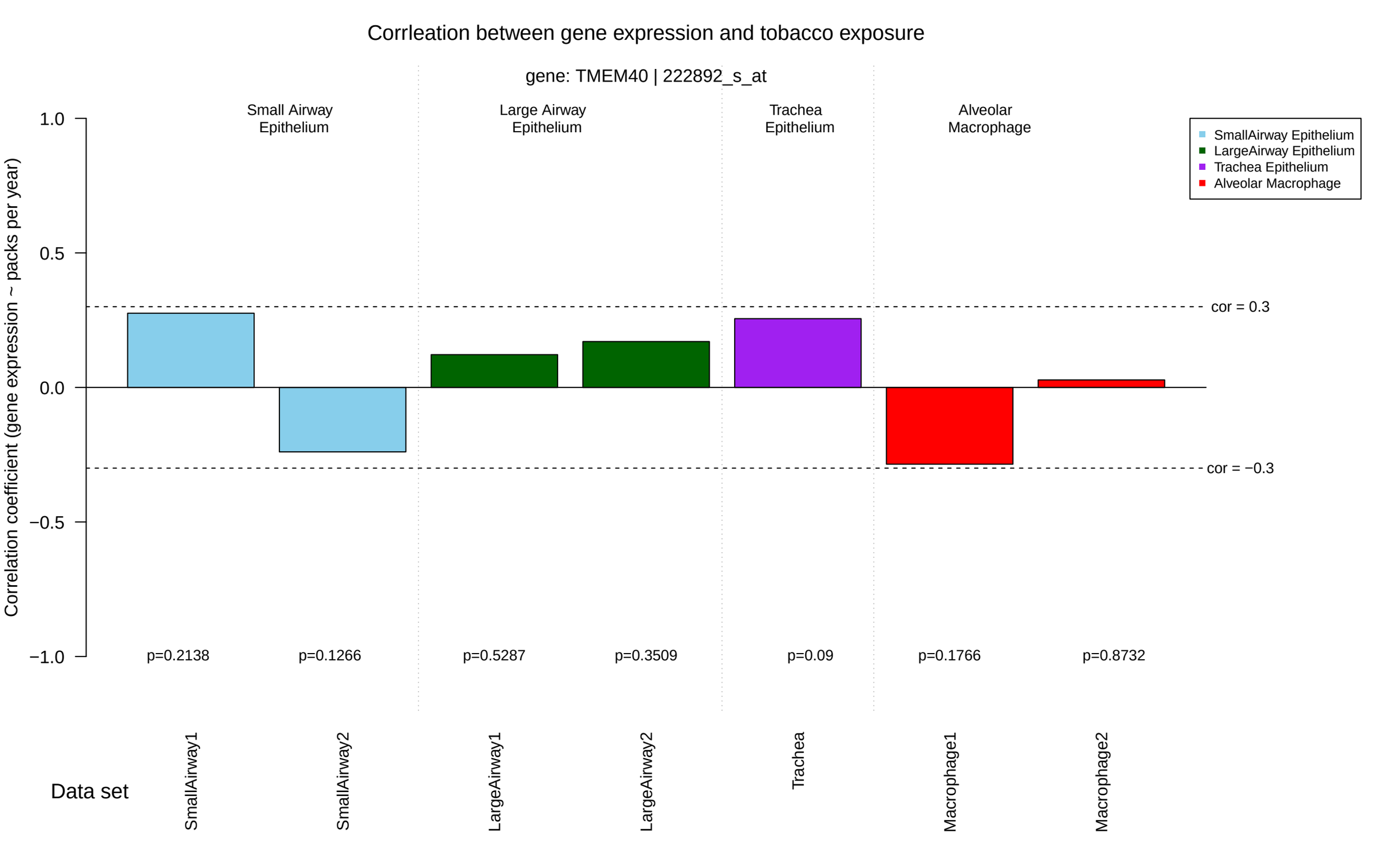

set: LargeAirway1 (p=0.9575)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2053)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0911)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3409)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4761)
TMEM40 | 222892_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM40 | 222892_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2138
p=0.1266
p=0.5287
p=0.3509
p=0.09
p=0.1766
p=0.8732
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2